Pooyan Jamshidi

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University of South Carolina
Computer Science and Engineering Department
Columbia, South Carolina 29208
pjamshid@cse.sc.edu
https://pooyanjamshidi.github.io

Research Interests

My research involves contributing to the theoretical foundations of Causal AI (e.g., structure learning, causal Bayesian optimization, causal representation learning) and Statistical ML (e.g., transfer learning, multiobjective optimization) and applying them to solve problems in Computer Systems (e.g., autonomous systems, robotics, ML inference pipelines, domain-specific AI accelerators, cloud-edge systems) and Sciences & Engineering (e.g., genomics, catalyst discovery).

Research Positions

8/2018–* **Assistant Professor (Computer Science & Engineering)**, *University of South Carolina*, Columbia, SC, US
Directing the AISys lab: https://pooyanjamshidi.github.io/AISys/

5/2021–12/2021 **Visiting Researcher**, *Google*, Mountain View, CA, US (remote)
Worked with Sugato Basu and Garima Pruthi in the AdsAI team (led by Sugato Basu) on Causal Representation Learning, ML Security, and Self-Supervised Learning.

12/2016–8/2018 **Postdoctoral Associate**, *Carnegie Mellon University*, Pittsburgh, US
Worked with Christian Kästner (postdoc advisor) on performance analysis of highly-configurable software, collaborated very closely with Norbert Siegmund. Worked with David Garlan on meta-learning for self-adaptive systems. Involved in BRASS MARS, a DARPA-sponsored project developing a model-based adaptation of mobile robotics.

2/2015–12/2016 **Postdoctoral Associate**, *Imperial College London*, London, UK
Involved in two EU projects (DICE and MODAClouds), where I developed methods and tools for auto-tuning big data systems (Apache Hadoop and Storm).

9/2014–2/2015 **Postdoc**, *Dublin City University*, Dublin, Ireland
Worked with Claus Pahl on developing self-learning controllers at IC4 cloud center and in collaboration with Intel (Giovani Estrada) and Microsoft (Niall Moran).

9/2010–9/2014 **Ph.D. Research Assistant**, *Dublin City University*, Dublin, Ireland
Worked with Claus Pahl (Ph.D. advisor) on developing a cloud controller for auto-scaling in the cloud. I received a scholarship from Lero (the Irish Software Research Centre).

7/2008–9/2010 **Research Team Lead**, *Shahid Beheshti University*, Tehran, Iran
Led the Automated Software Engineering team in a research lab (Lab Director: Dr. Fereidoon Shams); I have developed and contributed to the development of methods and tools for automating some processes in building distributed systems with service-oriented architectures.

Qualifications

9/2010–9/2014 **Ph.D., Computer Science**, *Dublin City University*, Ireland
- Thesis: *A framework for robust control of uncertainty in self-adaptive software*
- Adviser: Prof. Claus Pahl, External examiner: Prof. Pete Sawyer (Lancaster)

9/2003–2/2006 **M.Sc., Industrial Engineering (Systems)**, *Amirkabir University of Technology*, Iran
- Thesis: *An integrated knowledge-based system for product design support*
- Adviser: Dr. Saeed Mansour

9/1999–9/2003 **B.A., Math & Computer Science**, *Amirkabir University of Technology*, Iran

Experience in Industry

Post Ph.D.

5/2021–12/2021 **Visiting Researcher**, *Google*, Mountain View, CA, US (remote)
Worked with Sugato Basu and Garima Pruthi in the AdsAI team (led by Sugato Basu) on Causal Representation Learning, ML Security, and Self-Supervised Learning.

Pre Ph.D.

2007–2010 **Project Manager**, *Negaran Co*, Tehran, Iran
Managed a large-scale software project on developing a distributed software system automating business processes in a charity organization that has offices in many cities. My team included 20-30 software engineers and designers. Technologies: .NET, Microsoft SQL Server, SOA compatible stacks.

2006–2007 **Enterprise Architect**, *Negaran Co*, Tehran, Iran
Developed the enterprise architecture and prepared the ICT business planning using IBM's Business Systems Planning (BSP) and the Zachman Framework. Designed the software architecture of the core systems using an extended ADL in Visual Paradigm.

2003–2006 **Software Engineer**, *Rayan Pardaz Kavosh*, Tehran, Iran
Developed server-side software, and designed finance and automation software systems. Added several new features to the code base of a manufacturing automation system. Technologies: Visual C++, COM/CORBA, Socket programming

Honors and Awards

UofSC's Breakthrough Stars — University of South Carolina's 2022 **Breakthrough Stars Award.** `https://sc.edu/uofsc/posts/2022/06/breakthrough_star_pooyan_jamshidi.php`

Junior Researcher Award — **Junior Researcher Award** of Computer Science and Engineering Department, University of South Carolina, 2020.

Distinguished Reviewer — ACM TOSEM Board of **Distinguished Reviewers**, August 2020. `https://dl.acm.org/journal/tosem/distinguished-reviewers-board`

Competition — Selected as one of the two **finalists** at DCU for a fully funded research visit to IBM Research Brazil, 2014.

Competition — Thesis in 3 **national finalists**, "I bet you didn't know that software can adapt itself on-the-fly", Ireland, 2013, `https://goo.gl/igfKYC`

Ph.D. Fellowship — Awarded Lero Graduate School in Software Engineering (LGSSE) scholarship on the structured Ph.D. in Software Engineering, 2010-2013, **$85,000**.

Iranian University Entrance Exam — Ranked **21$^{\text{nd}}$** in the national graduate-level exam among 20,000 participants, 2003.

Media Coverage and Press Release

Podcast Interview	Dr. Alireza Dehghani, in his recent podcast, interviews me about my journey from undergraduate to becoming an assistant professor at UofSC, August 2022, https://www.youtube.com/watch?v=ZZarpnDKOY0
Interview with Bungee Pi Part 2	Learn math if you want to get into AI, June 2022, https://www.youtube.com/watch?v=-Fkvl8Oin8I
Interview with Bungee Pi Part 1	Breakthrough Ideas in Artificial Intelligence and Mathematics, June 2022, https://www.youtube.com/watch?v=YdwQXW2z1bs
Causal AI for Systems	Research intends to develop a shift in testing and debugging for modern machine learning systems, January 2022, https://www.sc.edu/study/colleges_schools/engineering_and_computing/news_events/news/2022/research_intends_to_develop_a_shift_in_testing_and_debugging_for_modern_machine_learning_systems.php
AI for Social Good	A new way to 'see' , June 2021, https://www.sc.edu/uofsc/posts/2021/06/smart_sight.php
AI in Space	USC researcher wants to train robots for NASA deep space missions, December 2020, https://www.postandcourier.com/columbia/news/usc-researcher-wants-to-train-robots-for-nasa-deep-space-missions/article_93d9bb3c-3afa-11eb-bd4c-7700ac496485.html
AI in Space	UofSC to develop AI-based autonomous systems for space missions, November 2020, https://www.sc.edu/study/colleges_schools/engineering_and_computing/news_events/news/2020/jamshidi_ai_space.php
AISys Lab	Media coverage of the AISys lab, June 2019, https://www.sc.edu/study/colleges_schools/engineering_and_computing/about/news/2019/jamshidiai.php

Funding

Roblox **"Unrestricted Gift"**, *Agency: Roblox Corporation*, Award Amount: **$20,000**, Fund Approved: May 2023
Role: PI. Collaborator: Tania Lorido-Botran (Roblox).

NSF **"Collaborative Research: EAGER: Towards a Design Methodology for Software-Driven Sustainability"**, *Agency: NSF*, Award Amount: **$300,000**, Project Period: 09/01/2022 - 08/31/2023
Role: Co-PI. PI: Eunsuk Kang (Carnegie Mellon University); other co-PIs: Mehdi Mirakhorli and Callie Babbitt (Rochester Institute of Technology).

NSF **"Collaborative Research: SHF: Medium: Causal Performance Debugging for Highly-Configurable Systems"**, *Agency: NSF*, Award Amount: **$1,200,000**, Project Period: 10/01/2021 - 09/30/2024
Role: PI. Co-PIs: Christian Kaestner (CMU) and Baishakhi Ray (Columbia).

NSF **"RTG: Mathematical Foundation of Data Science at University of South Carolina"**, *Agency: NSF*, Award Amount: **$1,996,609**, Project Period: 08/01/2021 - 07/31/2026
Role: Co-PI. Co-PIs (colleagues in the Math department): Wolfgang Dahmen, Linyuan Lu (PI), Wuchen Li, and Qi Wang.

NSF **"SmartSight: an AI-Based Computing Platform to Assist Blind and Visually Impaired People"**, *Agency: NSF*, Award Amount: **$499,650**, Project Period: 10/01/2020 - 10/01/2023
Role: PI. Co-PIs: Mohsen Amini Salehi (UL).

NASA **"RASPBERRY SI: Resource Adaptive Software Purpose-Built for Extraordinary Robotic Research Yields - Science Instruments"**, *Agency: NASA*, Award Amount: **$300,000**, Project Period: 10/01/2020 - 10/01/2023
Role: PI. Co-PIs: David Garlan (CMU, co-I), Bradley Schmerl (CMU, co-I), Javier Camara (York, collaborator), Ellen Czaplinski, Katherine Dzurilla (University of Arkansas, consultant), Matt DeMinico (NASA Glenn Research Center, consultant), Mike Dalal (NASA Ames, testbed), Hari Nayar (NASA JPL, testbed).

NASA **"A Generic Data-Driven Framework via Physics-Informed Deep Learning"**, *Agency: NASA*, Award Amount: **$100,000**, Project Period: 08/01/2020 - 08/01/2021
Role: Co-PI. Co-PIs: Lang Yuan (Mechanical Engineering).

Google Cloud **"GCP research credits for Adversarial ML"**, *Agency: Google*, Award Amount: **$10,000**, Project Period: 01/01/2020 - 06/01/2020
Role: PI.

Magellan Scholar Award **"Multi-stage Compression of Deep Neural Networks through Pruning and Knowledge Distillation"**, *Agency: UofSC Office of Undergraduate Research*, Award Amount: **$2,750**, Project Period: 01/01/2020 - 12/31/2020
Role: PI.

Magellan Scholar Award **"Ensemble of Many Weak Defenses: Defending Deep Neural Networks Against Adversarial Attacks"**, *Agency: UofSC Office of Undergraduate Research*, Award Amount: **$2,750**, Project Period: 01/01/2020 - 12/31/2020
Role: PI.

SURF **"Adversarial Machine Learning"**, *Agency: USC Honors College SURF Grant*, Award Amount: **$2,000**, Project Period: 10/15/2019 - 06/30/2020
Role: PI.

Google Cloud **"GCP research credits for Adversarial ML"**, *Agency: Google*, Award Amount: **$8,000**, Project Period: 09/01/2019 - 03/01/2020
Role: PI.

NASA **"Robust Software Testing of Autonomous Aerospace Robotic Systems Using Transfer Learning"**, *Agency: NASA*, Award Amount: **$50,000** (25,000 cost share), Project Period: 05/07/2019 - 05/06/2020
Role: PI; Co-PIs: Gregory Gay, Jason O'Kane.

ASPIRE-I **"Optimizing Energy Consumption of Deep Neural Networks for Intelligent Learning Systems"**, *Agency: University of South Carolina ASPIRE-I*, Award Amount: **$15,000**, Project Period: 07/01/2019 - 09/30/2020
Role: PI

DARPA-DOD-AFOSR **"Online Transfer Learning and Self-Adaptation of Robots"**, *Agency: Air Force Office of Science Research (AFOSR) and Defense Advanced Research Projects Agency (DARPA)*, Award Amount: **$114,741** (sub-award only), Project Period: 09/01/2018 - 08/31/2019
Role: PI (subaward); Co-PIs: Jonathan Aldrich (PI), David Garlan, Christian Kaestner, Claire Le Goues, and Manuela Veloso.

McNAIR **"Bayesian Structure Learning"**, *Agency: University of South Carolina McNair Junior Fellows*, Award Amount: **$3,000**, Project Period: 05/15/2019 - 08/16/2019
Role: PI.

SURF **"Neurofeedback-based Reinforcement Learning"**, *Agency: USC Honors College SURF Grant*, Award Amount: **$2,600**, Project Period: 10/15/2018 - 06/30/2019
Role: PI.

Invited Talks, Lectures and Seminars

Materials	The slides of my recent talks are available at: `https://pooyanjamshidi.github.io/talks/`
ICSE/SEAMS'23	**Learning from Valerie Issarny: Insights Gained from Program Co-Chairing SEAMS'23**, *Melbourne*, 2023
Oktoberbest	**Experiential Learning by Building Real-World AI Systems**, *Oktoberbest: A Symposium On Teaching*, 2023
NASA JPL	**Understanding and Explaining the Root Causes of Performance Faults with Causal AI: A Path towards Building Dependable Computer Systems**, *NASA JPL, Pasadena, CA*, 2022
exploreCSR	**Causal AI for Systems**, *exploreCSR: Democratizing AI*, 2022
IEEE SMDS	**Causal AI for Systems**, *Virtual*, 2021
Stanford MLSys	**Causal AI for Systems**, *Virtual*, 2021
Dagstuhl Seminar	**Causal Inference for Performance Analyses and Debugging of Serverless Systems**, *Virtual*, 2021
ASE Tutorial	**Machine Learning meets Software Performance**, *2021*
University of Louisiana	**ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense**, *Virtual*, November 2020
CMU	**ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense**, *Virtual*, September 2020
UPenn	**ATHENA: A Framework based on Diverse Weak Defenses for Building Adversarial Defense**, *Virtual*, September 2020
Google Headquarters	**Ensembles of Many Diverse Weak Defenses can be Strong: Defending Deep Neural Networks Against Adversarial Attacks**, *Virtual*, September 2020
AI Summit at Silicon Valley	**Robust Causal Transfer Learning**, *Virtual*, September 2020
Augusta University	**Ensembles of Many Diverse Weak Defenses can be Strong: Defending Deep Neural Networks Against Adversarial Attacks**, *Augusta, Georgia*, February 2020
Furman University	**Transfer Learning for Performance Analysis of Machine Learning Systems**, *Greenville, SC*, April 2019
SC EPSCoR Conference	**Transfer Learning for Performance Analysis of Machine Learning Systems**, *Greenville, SC*, April 2019
re-work DevOps Summit	**Machine Learning meets DevOps: Transfer Learning for Performance Optimization**, *Houston, Texas*, November 2018
SATURN	**Architectural Tradeoffs in Learning-Based Software**, *Plano, Texas*, May 2018
NC State University	**Learning Software Performance Models for Dynamic and Uncertain Environments**, *Raleigh, US*, 2017
SPEC DevOps RG	**An Exploratory Analysis of Transfer Learning for Performance Modeling of Configurable Systems**, *Online talk, RG DevOps Performance Working Group*, 2017

Dagstuhl Seminar	**Machine Learning meets DevOps**, *Software Performance Engineering in the DevOps World, Dagstuhl, Germany*, 2016
Bern University	**An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems**, *Bern, Switzerland*, 2016
SPEC DevOps RG	**An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems**, *Online talk, RG DevOps Performance Working Group*, 2016
SPEC DevOps RG	**Microservices Architecture Enables DevOps: Migration to a Cloud-Native Architecture**, *Online talk, RG DevOps Performance Working Group*, 2016
NC4 Conference	**DevOps: Migration to a Cloud-Native Architecture**, *The National Conference on Cloud Computing & Commerce, Dublin, Ireland*, 2016
Sharif University	**Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures**, *Tehran, Iran*, 2016
NII Shonan Meeting	**Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures**, *National Institute of Informatics (NII), Controlled Adaptation of Self-adaptive Systems (CASaS), Shonan, Japan*, 2016
Trinity College	**Self-learning Cloud Controllers**, *Trinity College Dublin*, 2015
UFC University	**Self-learning Cloud Controllers**, *Federal University of Ceará, Fortaleza, Brazil*, 2015
UECE University	**Cloud Migration Patterns: A Multi-Cloud Architectural Perspective**, *Ceará State University, Fortaleza, Brazil*, 2015
NC4 Conference	**Fuzzy Q-Learning for Knowledge Evolution**, *The National Conference on Cloud Computing & Commerce, Dublin, Ireland*, 2015
SPEC DevOps RG	**Self-learning Cloud Controllers**, *Online talk, RG DevOps Performance Working Group*, 2015
Dagstuhl Seminar	**Fuzzy Control Meets Software Engineering**, *Control Theory meets Software Engineering, Dagstuhl, Germany*, 2014

Teaching Activities

Teaching (Current)

Fall 2021-* **CSCE 212: Introduction to Computer Architecture**, *University of South Carolina, Columbia, SC*, Instructor, `https://pooyanjamshidi.github.io/csce212/`

Fall 2018-* **CSCE 585: Machine Learning Systems**, *University of South Carolina, Columbia, SC*, Instructor, `https://pooyanjamshidi.github.io/mls/`

Spring 2019-* **CSCE 580: Artificial Intelligence**, *University of South Carolina, Columbia, SC*, Instructor, `https://pooyanjamshidi.github.io/csce580/`

Teaching (Past)

Graduate Teaching

4/2018 **S17-655 Architectures for Software Systems (CMU Software Engineering Masters Program)**, *Carnegie Mellon University, Pittsburgh, US*, A guest lecture on Machine Learning for the Software Architect

11/2016 **SMA: Software Modeling and Analysis (Oscar Nierstrasz's course)**, *Bern University, Switzerland*, A guest lecture on Architecture Extraction

10/2015–1/2016 **424H - Learning in Autonomous Systems**, *Imperial College London*, TA

11/2014–12/2014 **CA674 - Cloud Architecture**, *Dublin City University*, Lectures shared with Claus Pahl

3/2014–6/2014 **CA668 E-commerce Infrastructure**, *Dublin City University*, Lectures shared with Claus Pahl

Undergraduate Teaching

10/2017 **Foundations of Software Engineering (Christian Kästner and Claire Le Goues)**, *Carnegie Mellon University*, Guest lecture on *Microservices*

9/2008-6/2010 **Software Engineering**, *Tarbiat Moallem University*, Lecturer

9/2001-6/2003 **Introduction to C/C++**, *Amirkabir University of Technology*, TA

9/2002-6/2003 **Data Structures**, *Amirkabir University of Technology*, TA

Postdocs

2022- **Sonam Kharde**

2022- **Mehdi Yaghouti**

2019-2022 **Mohammad Ali Javidian**

Doctoral Students

2018- **Shahriar Iqbal**, *Defended his proposal in January'23*

2023- **Abdolrasool Sharifi**, *Transferred to USC from the University of Virginia*

2021- **Abir Hossen**

2021- **Fatehmeh Ghofrani**

2021- **Hamed Damirchi**

2019-2022 **Jianhai Su**

2019-2022 **Ying Meng**

2022-2023 **Nasrin Imanpour**

2020-2021 **Shuge Lei**

PhD Interns

2022 **Saeid Ghafouri**, *Queen Mary University of London*

Undergraduate Students (Current)

4/2023- **Kartik Singhal**, *IIIT, Delhi, India*
1/2022- **Samuel Whidden**

Masters students (Completed)

2019-2020 **Peter Mourfield**

Undergraduate students (completed)

9/2020-4/2023 **Kimia Noorbakhsh**, *Now: PhD student at MIT*
3/2022-12/2022 **Hung-Tien Huang**
4/2020-6/2022 **Om Pandey**, *Now: Graduate student at Texas Austin*
4/2021-2/2022 **Ahana Biswas**, *Now: Graduate student at Pitt*
5/2021-8/2021 **Madelyn Khoury**, *REU*
5/2021-8/2021 **Bruce Brasseur**, *REU*
9/2020-4/2021 **Cody Shearer**, *RA*
8/2018-6/2019 **Nathan Stofik**, *RA*
12/2018-8/2019 **Tristan Klintworth**, *RA*
8/2019-5/2020 **Blake Edwards**, *RA*
8/2019-5/2020 **Stephen Baione**, *RA*
5/2019-8/2019 **Joshua Ravishankar**, *Summer REU*
5/2019-8/2019 **Rabina Phuyel**, *Summer REU*

Pre-college interns

6/2021- **Lane Stanley**, *Heathwood Hall High School, Columbia, SC*
2/2021- **Rohan Bafna**, *Riverside High School, Greenville, SC*
4/2021- **Rohit Rajagopalan**, *Southside High school, Greenville, SC*

Research Co-Mentoring

2023 **Carnegie Mellon University**, *REU Program*, Haesue Baik (first-year undergrad at University of Michigan, Ann Arbor), co-advised by Rohan Padhye, Vasudev Vikram (CMU PhD Student)
Understanding coverage guided fuzzing, understanding the performance of configurable systems, and designing custom fuzzing functions.

2022 **Carnegie Mellon University**, *REU Program*, Janhvi Somaiya (first-year undergrad at Rice University), co-advised by Christian Kaestner (CMU), Sonam Kharde (USC postdoc), Shahriar Iqbal (USC PhD student)
On identifying configuration-specific performance bugs with fuzzing.

2023 **Federal University of Minas Gerais (UFMG)**, *M.Sc. thesis*, Markos Viggiato de Almeida
On the Investigation of Software Development and Evolution Practices.

2017-2018 **Federal University of Minas Gerais (UFMG)**, *M.Sc. thesis*, Markos Viggiato de Almeida
On the Investigation of Software Development and Evolution Practices.

2018 **Carnegie Mellon University**, *Undergraduate research project*, Students: Alex Gao, Connor Lin, Jason Bak, Sander Lanbo Shi, Yunjie Su
Design space explorations of deep neural network architectures for embedded devices.

2017 **Carnegie Mellon University**, *REU Program*, Changming Xu, co-advised by Christian Kaestner (CMU)
Can you fool a self-adaptive software system?

2016–17 **Imperial College London**, *B.Eng. project*, Ka Yan Wong
Experimental study of performance variations in big data systems.

2016 **Imperial College London**, *M.Eng. project*, Yifan Zhai
A DevOps canary testbed for Big Data application testing.

2015 **Imperial College London**, *B.Eng. final project*, Zhang Haoran, Qiu Jiaxin, Abdeljallal Fahd, Lu Cong, Chadjiminas Ioannis, Liu Yao
A suite for automated configuration testing and benchmarking for Apache Spark.

2016 **Imperial College London**, *M.Eng. final project*, Xidi Chen
A suite for automated configuration testing and benchmarking for Apache Hadoop.

2014–2015 **Dublin City University**, *M.Sc. practicum*, Robert Mason
Auto-scaling in OpenStack cloud.

2014 **Dublin City University**, *MS.c. practicum*, Brian C. Carroll
Auto-scaling in the cloud: evaluating a control-based technique.

2014–15 **Sharif University of Technology**, *M.Sc. thesis*, Armin Balalaie
Migrating to cloud-native architectures using microservices.

2008–10 **Shahid Beheshti University**, *M.Sc. thesis*, Ali Rostampour, Ali Kazemi
A metric for measuring the degree of entity-centric service cohesion.

PhD Committee Member

2023 **Mohammed E. Elbtity**, *Advisor: Ramtin Zand*, University of South Carolina, Computer Science and Engineering
Approximate Computing and In-Memory Computing: The best of the two worlds!

2023 **Bharat Joshi**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
Robust Underwater State Estimation and Mapping

2023 **Luc Lesoil**, *Advisor: Mathieu Acher*, INSA Rennes, France, Deep Software Variability for Resilient Performance Models of Configurable Systems

2022 **Manas Gaur**, *Advisor: Amit Sheth*, University of South Carolina, Computer Science and Engineering
Kowledge-Infused Learning

2022 **Aron Hein**, *Advisor: Homayoun Valafar*, University of South Carolina, Computer Science and Engineering

2022 **Olajide H. Bamidele**, *Advisor: Andreas Heyden*, University of South Carolina, Chemical Engineering

2021 **Rasika Jayarathna**, *Advisor: Jochen Lauterbach*, University of South Carolina, Chemical Engineering

2021 **Alireza Salahirad**, *Advisor: Greg Gay*, University of South Carolina, Computer Science and Engineering
Empirical Studies on Automated Software Testing Practices

2020 **Elizabeth Stewart**, *Advisor: Brett Sherman*, University of South Carolina, Philosophy
Alexa, Should I Trust You? A Theory of Trustworthiness for Artificial Intelligence

2020 **Nare Karapetyan**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
Area Coverage Path Planning Problem in Aquatic Environments

2020 **Hazhar Rahmani**, *Advisor: Jason O'Kane*, University of South Carolina, Computer Science and Engineering
Automata theoretic approaches to planning in robotics: combinatorial filter minimization, planning to chronicle, and temporal logic planning with soft specifications

2020 **William Hoskins**, *Advisor:*, University of South Carolina, Computer Science and Engineering

2019 **Mohammad Ali Javidian**, *Advisor: Marco Valtorta*, University of South Carolina, Computer Science and Engineering
Advanced Topics in Probabilistic Graphical Models: Properties, Learning Algorithms, and Applications

2019 **Hussein Almulla**, *Advisor: Greg Gay*, University of South Carolina, Computer Science and Engineering
Learning How to Search: Generating Effective Test Cases Through Adaptive Fitness Function Selection

2019 **Shervin Ghasemlou**, *Advisor: Jason O'Kane*, University of South Carolina, Computer Science and Engineering
Algorithmic Robot Design: Label Maps, Procrustean Graphs, and the Boundary of Non-destructiveness

2018 **Hayder Dawood Abbood**, *Advisor: Andrea Benigni*, University of South Carolina, Electrical Engineering
Data-Driven Modeling Through Power Hardware in the Loop Experiments: A PV Micro-Inverter Example

2018 **Jason Moulton**, *Advisor: Ioannis Rekleitis*, University of South Carolina, Computer Science and Engineering
A Novel and Inexpensive Solution to Build Autonomous Surface Vehicles Capable of Negotiating Highly Disturbed Environments

M.S. Committee Member

2020 **Noah Geveke**, *Advisr: Marco Valtorta*, University of South Carolina, Computer Science and Engineering
On the Robustness of Bayesian Network Learning Algorithms against Malicious Attacks

Publications

Key publications are highlighted with ★ based on their importance to my research goal, http://scholar.google.com/citations?user=41rV5koAAAAJ

Refereed Journal and Top Magazine Articles

★JSys [J30] S. Ghafouri, K. Razavi, M. Salmani, A. Sanaee, T. Lorido-Botran, L. Wang, J. Doyle, P. Jamshidi, *IPA: Inference Pipeline Adaptation to Achieve High Accuracy and Cost-Efficiency*, Journal of Systems Research, 2023.

★RA-L [J29] A. Hossen, S. Kharade, B. Schmerl, J. Cámara, J. M. O'Kane, E. C. Czaplinski, K. A. Dzurilla, D. Garlan, P. Jamshidi, *CaRE: Finding Root Causes of Configuration Issues in Highly-Configurable Robots*, IEEE Robotics and Automation Letters (RA-L), 2023 (Also appearing at IROS'23).

★JAIR [J28] S. Iqbal, J. Su, L. Kotthoff, P. Jamshidi, *FlexiBO: Cost-Aware Multi-Objective Optimization of Deep Neural Networks*, Journal of Artificial Intelligence Research (JAIR), 2023.

★JAIR [J27] M. Javidian, M. Valtorta, P. Jamshidi, *AMP Chain Graphs: Minimal Separators and Structure Learning Algorithms*, Journal of Artificial Intelligence Research (JAIR), 2020. doi:10.1613/jair.1.12101 [SJR rating: **Q1**]

PCCP [J26] K. McCullough, T. Williams, K. Mingle, P. Jamshidi, J. Lauterbach, *High-throughput Experimentation meets Artificial Intelligence: A New Pathway to Catalyst Discovery*, Physical Chemistry Chemical Physics (PCCP), 2020. doi:10.1039/D0CP00972E [SJR rating: **Q1**]

★Springer ASE [J25] M. Velez, P. Jamshidi, F. Sattler, N. Siegmund, S. Apel, C. Kaestner, *ConfigCrusher: White-Box Performance Analysis for Configurable Systems*, Springer Automated Software Engineering, 2020. doi:10.1007/s10515-020-00273-8 [SJR rating: **Q1**]

IEEE TSE [J24] R. Krishna, V. Nair, P. Jamshidi, T. Menzies, *Whence to Learn? Transferring Knowledge in Configurable Systems using BEETLE*, IEEE Transactions on Software Engineering (TSE), 2020. doi:10.1109/TSE.2020.2983927 [SJR rating: **Q1**]

★Software [J23] N. C Mendonça, P. Jamshidi, D. Garlan, and C. Pahl, *Developing Self-Adaptive Microservice Systems: Challenges and Directions*, IEEE Software, 2019. [SJR rating: **Q1**]

Software [J22] J. Aldrich, J. Biswas, J. Camára, D. Garlan, A. Guha, J. Holtz, P. Jamshidi, C. Kaestner, C. Le Goues, A. Mohseni-Kabir, I. Ruchkin, S. Samuel, B. Schmerl, C. Steven Timperley, M. Veloso, and I. Voysey, *Model-based Adaptation for Robotics Software*, IEEE Software, 2019. [SJR rating: **Q1**]

Springer JBD [J21] M. Bersani, F. Marconi, D. Tamburri, A. Nodari, P. Jamshidi, *Verifying big data topologies by-design: a semi-automated approach*, Journal of Big Data, 2019. doi:10.1186/s40537-019-0199-y [SJR rating: **Q1**]

Software [J20] C. Trubiani, P. Jamshidi, J. Cito, W. Shang, Z.M. Jiang, M. Borg, *Performance issues? Hey DevOps, mind the uncertainty!*, IEEE Software, 2018. [SJR rating: **Q1**]

Software [J19] **Featured Article** P. Jamshidi, C. Pahl, N. Mendonça, J. Lewis, S. Tilkov, *Microservices: The Journey So Far and Challenges Ahead*, IEEE Software, 2018. doi:10.1109/MS.2018.2141039, [SJR rating: **Q1**]

Wiley SPE [J18] A. Balalaie, A. Heydarnoori, P. Jamshidi, D.A. Tamburri, T. Lynn, *Microservices migration patterns*, Wiley Software: Practice and Experience (SPE), 2018. doi:10.1002/spe.2608 [SJR rating: **Q2**]

Elsevier JSS [J17] A. Aleti, C. Trubiani, A. van Hoorn, P. Jamshidi, *An Efficient Method for Uncertainty Propagation in Robust Software Performance Estimation*, Elsevier Journal of Systems and Software (JSS), 2018. doi:10.1016/j.jss.2018.01.010 [SJR rating: **Q1**]

ACM TOIT [J16] C. Pahl, P. Jamshidi, O. Zimmermann, *Architectural Principles for Cloud Software*, ACM Transactions on Internet Technology (TOIT), 18(2), 2018. doi:10.1145/3104028 [SJR rating: **Q1**]

IEEE TCC [J15] C. Pahl, A. Brogi, J. Soldani, P. Jamshidi, *Cloud Container Technologies: a State-of-the-Art Review*, IEEE Transactions on Cloud Computing (TCC). 2017 doi:10.1109/TCC.2017.2702586. [SJR rating: **Q1**]

Wiley JSEP [J14] C Pahl, P. Jamshidi, D Weyns, *Cloud architecture continuity: Change models and change rules for sustainable cloud software architectures*, Wiley Journal of Software: Evolution and Process (JSEP), 29(2), 2017 doi:10.1002/smr.1849. [SJR rating: **Q2**]

ACM TAAS [J13] **Invited** A. Filieri, M. Maggio, K. Angelopoulos, N. D'Ippolito, I. Gerostathopoulos, A. Hempel, H. Hoffmann, P. Jamshidi, E. Kalyvianaki, C. Klein, F. Krikava, S. Misailovic, A. V. Papadopoulos, S. Ray, A. M. Sharifloo, S. Shevtsov, M. Ujma and T. Vogel, *Control Strategies for Self-Adaptive Software Systems*, ACM Transactions on Autonomous and Adaptive Systems (TAAS), invited paper, 11(4), 2017, doi:10.1145/3024188. [SJR rating: **Q1**]

Cloud [J12] P. Jamshidi, C. Pahl, N. Mendonça, *Managing Uncertainty in Autonomic Cloud Elasticity Controllers*, IEEE Cloud Computing, 2016. doi:10.1109/MCC.2016.66

Software [J11] **Featured Article** A. Balalaie, A. Heydarnoori, P. Jamshidi, *Microservices Enables DevOps: an Experience Report on Migration to a Cloud-Native Architecture*, IEEE Software, 2016. doi:10.1109/MS.2016.64, [SJR rating: **Q1**]

IEEE TCC [J10] F. Fowley, C. Pahl, P. Jamshidi, D. Fang, X. Liu, *A Classification and Comparison Framework for Cloud Service Brokerage Architectures*, IEEE Transactions on Cloud Computing (TCC), 2016, doi:10.1109/TCC.2016.2537333. [SJR rating: **Q1**]

Wiley SPE [J9] P. Jamshidi, C. Pahl, N. C. Mendonça, *Pattern-based Multi-Cloud Architecture Migration*, Wiley Software: Practice and Experience (SPE), 47(9), 1159-1184, 2016. doi:10.1002/spe.2442 [SJR rating: **Q2**]

★ Elsevier FGCS [J8] S. Farokhi, P. Jamshidi, E. B. Lakew, I. Brandic, E. Elmroth, *A Hybrid Cloud Controller for Vertical Memory Elasticity: A Control-theoretic Approach*, Elsevier Future Generation Computer Systems (FGCS), 65, 57 – 72 (2016). doi:10.1016/j.future.2016.05.028, [SJR rating: **Q1**]

Elsevier FGCS [J7] D. Fang, X. Liu, I. Romdhani, P. Jamshidi, C. Pahl, *An Agility-Oriented and Fuzziness-Embedded Semantic Model for Collaborative Cloud Service Search, Retrieval and Recommendation*, Elsevier Future Generation Computer Systems (FGCS), 56, 11 – 26 (2016). doi:10.1016/j.future.2015.09.025, [SJR rating: **Q1**]

IEEE TCC [J6] **Featured Article** P. Jamshidi, A. Ahmad, C. Pahl, *Cloud Migration Research: A Systematic Review*, IEEE Transactions on Cloud Computing (TCC), 1(2), 142 – 157 (2013). doi:10.1109/TCC.2013.10, [SJR rating: **Q1**]

Springer JSEP [J5] A. Ahmad, P. Jamshidi, C. Pahl, *Classification and Comparison of Architecture Evolution Reuse Knowledge - A Systematic Review*, Springer Journal of Software: Evolution and Process (JSEP), 26(7): 654–691 (2014). doi:10.1002/smr.1643, [SJR rating: **Q2**]

EASST [J4] A. Ahmad, P. Jamshidi, C. Pahl, F. Khaliq, *A Pattern Language for the Evolution of Component-based Software Architectures*, Electronic Communications of the EASST, 59, 1 – 32 (2014). doi:10.14279/tuj.eceasst.59.931

IEEE Systems [J3] A. Khoshkbarforoushha, P. Jamshidi, M. Fahmideh, L. Wang, R. Ranjan, *Metrics for BPEL Process Reusability Analysis in a Workflow System*, IEEE Systems Journal, 1 – 10 (2014). doi:10.1109/JSYST.2014.2317310, [SJR rating: **Q1**]

Springer SoSym [J2] M. Fahmideh, M. Sharifi, P. Jamshidi, *Enhancing the OPEN Process Framework with Service-Oriented Method Fragments*, Springer Software and Systems Modeling (SoSym), 13(1): 361 – 390 (2014). doi:10.1007/s10270-011-0222-z, [SJR rating: **Q1**]

Springer SOCA [J1] A. Khoshkbarforoushha, P. Jamshidi, A. Nikravesh, F. Shams, *Metrics for BPEL process context-independency analysis*, Springer Service Oriented Computing and Applications (SOCA), 5(3): 139 – 157 (2011). doi:10.1007/s11761-011-0077-8, [SJR rating: **Q2**]

Refereed Conference Papers

★SoCC [C45] M.S. Iqbal, Z. Zhong, I. Ahmad, B. Ray, P. Jamshidi, *CAMEO: A Causal Transfer Learning Approach for Performance Optimization of Configurable Computer Systems.*, In Proc. of ACM Symposium on Cloud Computing (SoCC'23), 2023.

RAP4Robots [C44] H. Damirchi, F. Agostinelli, P. Jamshidi, *Independent Modular Networks*, In Proc. of Workshop on Effective Representations, Abstractions, and Priors for Robot Learning (RAP4Robots at ICRA'23), 2023.

AdvML-Frontiers [C43] F. Ghofrani, M. Yaghouti, P. Jamshidi, *Rethinking Robust Contrastive Learning from the Adversarial Perspective*, In Proc. of the 2nd New Frontiers in Adversarial Machine Learning (AdvML Frontiers at ICML'23), 2023.

★EuroMLSys [C42] M. Salmani, S. Ghafouri, A. Sanaee, K. Razavi, M. Mühlhäuser, J. Doyle, P. Jamshidi, M. Sharifi, *Reconciling High Accuracy, Cost-Efficiency, and Low Latency of Inference Serving Systems*, In Proc. of Proceedings of the 3rd Workshop on Machine Learning and Systems (EuroMLSys), 2023.

CLOUD [C41] A. Mokhtari, A. Hossen, P. Jamshidi, M. Amini Salehi, *FELARE: Fair Scheduling of Machine Learning Applications on Heterogeneous Edge Systems*, In Proc. of International Conference on Cloud Computing (IEEE CLOUD), 2022 [Acceptance rate: 20%].

★AutoML [C40] S. Iqbal, J. Su, L. Kotthoff, P. Jamshidi, *Getting the Best Bang For Your Buck: Choosing What to Evaluate for Faster Bayesian Optimization*, International Conference on Automated Machine Learning (AutoML), 2022.

★EuroSys [C39] S. Iqbal, R. Krishna, M.A. Javidian, B. Ray, P. Jamshidi, *Unicorn: Reasoning about Configurable System Performance through the lens of Causality*, In Proc. of European Conference on Computer Systems (EuroSys), 2022 [Acceptance rate: 25%(42/162)].

★Arxiv [C38] Kimia Noorbakhsh, Modar Sulaiman, Mahdi Sharifi, Kallol Roy, P. Jamshidi, *Pretrained Language Models are Symbolic Mathematics Solvers too!*, [under review, available on arxiv].

★ICSE [C37] M. Velez, P. Jamshidi, N. Siegmund, S. Apel, C. Kaestner, *On Debugging the Performance of Configurable Software Systems: Developer Needs and Tailored Tool Support*, In Proc. of International Conference on Software Engineering (ICSE), 2022 [Acceptance rate: 26%(197/751)].

★NeurIPS WHY [C36] M.A. Javidian, O. Pandey, P. Jamshidi, *Scalable Causal Domain Adaptation*, In Proc. NeurIPS WHY-21 (Causal Inference and Machine Learning: Why now?), 2021 [Invited for oral presentation; Acceptance rate: 8%(4/50)].

★ICSE [C35] M. Velez, P. Jamshidi, N. Siegmund, S. Apel, C. Kaestner, *White-Box Analysis over Machine Learning: Modeling Performance of Configurable Systems*, In Proc. of International Conference on Software Engineering (ICSE), Virtual (May 2021) [Acceptance rate: 23%(138/602)].

AAMAS [C34] M. Rahman, A. Rasheed, M. Khan, M.A. Javidian, P. Jamshidi, Md. Mamun-Or-Rashid, *Accelerating Recursive Partition-Based Causal Structure Learning Using An Improved Structure Refinement Approach*, In Proc. of International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Virtual (May 2021) [Acceptance rate: 25%(152/612)].

NeurIPS ML for Systems [C33] S. Iqbal, R. Krishna, M.A. Javidian, B. Ray, P. Jamshidi, *CADET: A Systematic Method For Debugging Misconfigurations using Counterfactual Reasoning*, In Proc. of NeurIPS ML for Systems, Virtual (December 2020).

PROFES [C32] A. Banijamali, P. Kuvaja, M. Oivo, P. Jamshidi, *Kuksa*: Self-adaptive Microservices in Automotive Systems*, In Proc. of International Conference on Product-Focused Software Process Improvement (PROFES), Virtual (November 2020).

★UAI [C31] M. Javidian, M. Valtorta, P. Jamshidi, *Learning LWF Chain Graphs: A Markov Blanket Discovery Approach*, In Proc. of The Conference on Uncertainty in Artificial Intelligence (UAI), Toronto, Canada (August 2020) [Acceptance rate: 27%(142/515)].

SUM [C30] M. Javidian, M. Valtorta, P. Jamshidi, *Order-Independent Structure Learning of Multivariate Regression Chain Graphs*, In Proc. of International Conference on Scalable Uncertainty Management (SUM), Compiegne, France (December 2019).

PROFES [C29] A. Banijamali, P. Jamshidi, P. Kuvaja, and M. Oivo, *Kuksa: A Cloud-Native Architecture for Enabling Continuous Delivery in the Automotive Domain*, In Proc. of International Conference on Product-Focused Software Process Improvement (PROFES), Barcelona, Spain (November 2019).

ICGSE [C28] M. Viggiato, J. Oliveira, E. Figueiredo, P. Jamshidi, and C. Kaestner, *Understanding similarities and differences in software development practices across domains*, In Proc. of International Conference on Global Software Engineering (ICGSE), Montreal, Canada, (May 2019).

ICPE [C27] C. Bezemer, S. Eismann, V. Ferme, J. Grohmann, R. Heinrich, P. Jamshidi, W. Shang, A. van Hoorn, M. Villavicencio, J. Walter, and F. Willnecker, *How is Performance Addressed in DevOps?*, In Proc. of International Conference on Performance Engineering (ICPE), Mumbai, India, (April 2019).

OpML [C26] M. S. Iqbal, L. Kotthoff, P. Jamshidi, *Transfer Learning for Performance Modeling of Deep Neural Network Systems*, In Proc. of the USENIX Conference on Operational Machine Learning (OpML), Santa Clara, CA, (May 2019).

★SEAMS [C25] P. Jamshidi, J. Camára, B. Schmerl, C. Kästner, D. Garlan, *Machine Learning Meets Quantitative Planning: Enabling Self-Adaptation in Autonomous Robots*, In Proc. of the 12th International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), Montreal, Canada, (May 2019) [Acceptance rate: 28%(10/36)].

AAAI [C24] M. A. Javidian, P. Jamshidi, M. Valtorta, *Transfer Learning for Performance Modeling of Configurable Systems: A Causal Analysis*, In Proc. of AAAI Spring Symposium Beyond Curve Fitting: Causation, Counterfactuals, and Imagination-based AI, Stanford, CA, USA, (March 2019).

AAMAS [C23] M. A. Javidian, P. Jamshidi, R. Ramezanian, *Avoiding Social Disappointment in Elections*, In Proc. of International Conference on Autonomous Agents and Multiagent Systems (AAMAS), Montreal, Canada, (May 2019).

FSE [C22] P. Jamshidi, M. Velez, C. Kästner, N. Siegmund, *Learning to Sample: Exploiting Similarities Across Environments to Learn Performance Models for Configurable Systems*, In Proc. of the ACM SIGSOFT Symposium on the Foundations of Software Engineering (FSE), Florida, USA, (Nov 2018) [Acceptance rate: 19%(55/295); CORE rating: rank **A***].

TechDebt [C21] A. Mori, G. Vale, M. Viggiato, J. Oliveira, E. Figueiredo, E. Cirilo, P. Jamshidi, and C. Kästner, *Evaluating Domain-Specific Metric Thresholds: An Empirical Study*, In Proc. of the International Conference on Technical Debt (TechDebt), Gothenburg, Sweden, (May 27-28, 2018).

★ASE [C20] P. Jamshidi, N. Siegmund, M. Velez, C. Kästner, A. Patel, Y. Agarwal, *Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis*, In Proc. of the 32nd IEEE/ACM International Conference on Automated Software Engineering (ASE), Illinois, USA, (Nov 2017) [Acceptance rate: 21%(67/322); CORE rating: rank **A***].

★SEAMS [C19] P. Jamshidi, M. Velez, C. Kästner, N. Siegmund, P. Kawthekar, *Transfer Learning for Improving Model Predictions in Highly Configurable Software*, In Proc. of the 12th International Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), Buenos Aires, Argentina, (May 2017) [Acceptance rate: 23% (14/61), Invited for an extension to ACM TAAS].

CCGrid [C18] H. Arabnejad, C. Pahl, P. Jamshidi, G. Estrada, *A Comparison of Reinforcement Learning Techniques for Fuzzy Cloud Auto-Scaling*, in Proc. of The 17th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid), Madrid, Spain, (May 2017) [Acceptance rate: 23% (64/280); CORE rating: rank **A**]. Nominated for the best paper award.

WICSA [C17] M. Bersani, F. Marconi, D. Tamburri, P. Jamshidi, A. Nodari, *Continuous Architecting of Stream-Based Systems*, In Proc. of The 13th Working IEEE/IFIP Conference on Software Architecture (WICSA), Venice, Italy, (April 2016). [Acceptance rate: 37% (56/149); CORE rating: rank **A**]

★MASCOTS [C16] P. Jamshidi, G. Casale, *An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems*, In Proc. of IEEE 24th International Symposium on Modeling, Analysis and Simulation of Computer and Telecommunication Systems (MASCOTS), London, UK (September 2016). [Acceptance rate: 17% (34/200); CORE rating: rank **A**]

★QoSA [C15] P. Jamshidi, A. Sharifloo, C. Pahl, H. Arabnejad, A. Metzger, G. Estrada, *Fuzzy Self-Learning Controllers for Elasticity Management in Dynamic Cloud Architectures*, In Proc. of 12th International ACM SIGSOFT Conference on the Quality of Software Architectures (QoSA), Venice, Italy, (April 2016). [CORE rating: rank **A**]

ICCAC [C14] P. Jamshidi, A. Sharifloo, C. Pahl, A. Metzger, G. Estrada, *Self-Learning Cloud Controllers: Fuzzy Q-Learning for Knowledge Evolution*, In Proc. of IEEE International Conference on Cloud and Autonomic Computing (ICCAC), Boston, MA, USA, (Sept. 2015). [CORE rating: rank B]

ICAC [C13] Soodeh Farokhi, P. Jamshidi, D. Lucanin, I. Brandic, *Performance-Based Vertical Memory Elasticity*, In Proc. of IEEE International Conference on Autonomic Computing (ICAC), Grenoble, France, (Jul. 2015). [CORE rating: rank B]

ECSA [C12] C. Pahl, P. Jamshidi, *Software Architecture for the Cloud - A Roadmap Towards Control-Theoretic, Model-Based Cloud Architecture*, In Proc. of Springer European Conference on Software Architecture (ECSA), (Sept. 2015). [CORE rating: rank **A**]

SEAMS [C11] **Invited** A. Filieri, M. Maggio, K. Angelopoulos, N. D'Ippolito, I. Gerostathopoulos, A. Hempel, H. Hoffmann, P. Jamshidi, E. Kalyvianaki, C. Klein, F. Krikava, S. Misailovic, A. V. Papadopoulos, S. Ray, A. M. Sharifloo, S. Shevtsov, M. Ujma and T. Vogel, *Software Engineering Meets Control Theory*, In Proc. of the 10th ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Firenze, Italy, (May 2015), [Acceptance rate: 29% (16/55)].

UCC [C10] L. Zhang, Y. Zhang, P. Jamshidi, L. Xu, C. Pahl, *Workload Patterns for Quality-Driven Dynamic Cloud Service Configuration and Auto-Scaling*, In Proc. of IEEE/ACM 7th International Conference on Utility and Cloud Computing (UCC), London, UK, (Dec 2014), [Acceptance rate: 19% (38/198); CORE rating: rank **A**]

SEAMS [C9] P. Jamshidi, A. Ahmad, C. Pahl, *Autonomic Resource Provisioning for Cloud-Based Software*, In Proc. of the 9th ACM International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Hyderabad, India, (Jun. 2014), [Acceptance rate= 18% (15/80)].

CSMR [C8] P. Jamshidi, M. Ghafari, A. Ahmad, C. Pahl, *A Framework for Classifying and Comparing Architecture-Centric Software Evolution Research*, In Proc. of 17th European Conference on Software Maintenance and Reengineering (CSMR), Genova, Italy, (Mar. 2013), [Acceptance rate: 36% (29/80); CORE rating: rank B]

CBSE [C7] M. Ghafari, P. Jamshidi, S. Shahbazi, H. Haghighi, *An architectural approach to ensure globally consistent dynamic reconfiguration of component-based systems*, In Proc. of the 15th ACM SIGSOFT symposium on Component-Based Software Engineering (CBSE), Bertinoro, Ital, (Sept. 2012). [Acceptance rate: 29%; CORE rating: rank **A**]

CAiSE [C6] A. Ahmad, P. Jamshidi, C. Pahl, *Graph-Based Pattern Identification from Architecture Change Logs*, In Proc. of Springer International Conference on Advanced Information Systems Engineering (CAiSE), (Jun. 2012). [Short Paper, CORE rating: rank **A**]

QSIC [C5] A. Kazemi, A. Rostampour, A. Zamiri, P. Jamshidi, H. Haghighi, F. Shams, *An Information Retrieval Based Approach for Measuring Service Conceptual Cohesion*, In Proc. of 11th IEEE International Conference on Quality Software (QSIC), Madrid, Spain, (Jul. 2011). [Acceptance rate: 17.6%; CORE rating: rank B]

SCC [C4] A. Kazemi, A. Nasirzadeh, A. Rostampour, H. Haghighi, P. Jamshidi, F. Shams, *Measuring the Conceptual Coupling of Services Using Latent Semantic Indexing*, In Proc. of IEEE International Conference on Services Computing (SCC), Washington, DC, USA, (Jul. 2011). [Acceptance rate: 17%; CORE rating: rank **A**]

SERVICES [C3] A. Kazemi, A. Rostampour, P. Jamshidi, E. Nazemi, F. Shams, A. Nasirzadeh, *A Genetic Algorithm Based Approach to Service Identification*, In Proc. of IEEE World Congress on Services (SERVICES), Washington, DC, USA, (Jul. 2011). [Acceptance rate: 17%; CORE rating: rank A]

SERVICES [C2] A. Khoshkbarforoushha, R. Tabein, P. Jamshidi, F. Shams, *Towards a metrics suite for measuring composite service granularity level appropriateness*, In Proc. of IEEE World Congress on Services (SERVICES), Miami, FL, USA, (Jul. 2010). [Acceptance rate: 18% (29/165); CORE rating: rank B]

SCC [C1] P. Jamshidi, M. Sharifi, S. Mansour, *To Establish Enterprise Service Model from Enterprise Business Model*, in Proc. of IEEE International Conference on Services Computing (SCC), Honolulu, HI, USA, (Jul. 2008). [Acceptance rate: 18%; CORE rating: rank **A**]

Worshop Papers and Technical Reports

Arxiv [TR4] AM. Roth, N. Topin, P. Jamshidi, M. Veloso, *Conservative Q-improvement: Reinforcement learning for an interpretable decision-tree policy*, Arxiv, (2019).

FC [TR3] S. Farokhi, P. Jamshidi, I. Brandic, E. Elmroth, *Self-adaptation Challenges for Cloud-based Applications: A Control Theoretic Perspective*, International Workshop on Feedback Computing, (2015).

Dagstuhl [TR2] A. van Hoorn, P. Jamshidi, P. Leitner, I. Weber, *Software Performance Engineering in the DevOps World*, Report from GI-Dagstuhl Seminar 16394, (Sept. 2017), `https://arxiv.org/abs/1709.08951`.

SPEC [TR1] A. Brunnert, A. van Hoorn, F. Willnecker, A. Danciu, Wi. Hasselbring, C. Heger, N. Herbst, P. Jamshidi, R. Jung, J. von Kistowski, A. Koziolek, J. Kroß, S. Spinner, C. Vögele, J. Walter, A. Wert, *Performance-oriented DevOps: A Research Agenda*, SPEC Research Group — DevOps Performance Working Group, Standard Performance Evaluation Corporation (SPEC), (Aug. 2015), SPEC-RG-2015-01.

Software Artifacts

Github Almost all listed software is developed collaboratively. LD: lead developer; CC: contributor.

CC [S17] **Unicorn**, *Unicorn is a framework for reasoning about system performance using Causal AI*, Python, https://github.com/softsys4ai/unicorn

★CC [S17] **ATHENA**, *is a Framework for defending machine learning systems against adversarial attacks*, Python, https://github.com/softsys4ai/athena

LD [S16] **robot_control**, *robot_control is a set of controllers and actuators that run, control, and interface the ROS-enabled robots, as a part of DARPA BRASS project.*, C++, https://github.com/pooyanjamshidi/robot_control

LD [S15] **brass_gazebo_battery**, *brass_gazebo_battery is a Gazebo plugin that simulates an open-circuit battery model. This is a fairly extensible and reusable battery plugin for any kind of Gazebo-compatible robots.*, C++, https://github.com/pooyanjamshidi/brass_gazebo_battery

LD [S14] **GenPerf**, *GenPerf uses symbolic regression to synthetically generate target performance influence models with different similarities to the source model. GenPerf is used to generate synthetic data for evaluating our TL approach.*, Python, https://github.com/pooyanjamshidi/GenPerf

LD [S13] **AutoTL (NOT RELEASED)**, *This tool enables an adaptive sampling that learns from multiple exclusive information origins including influential configuration options, their interactions, and performance distribution of the configurable software*, Python, https://github.com/pooyanjamshidi/autotl

LD [S12] **model-learner**, *This tool enables discovering a black box model using regression models and transfer learning. This was used in the BRASS project to enable battery charge/recharge in a self-adaptive loop*, Python, https://github.com/cmu-mars/model-learner

LD [S11] **autoscaling-bigdata**, *A library for application level runtime monitoring and runtime change actuators and auto-scaling controllers for Big Data technologies such as Apache Storm, Spark, Hadoop*, Matlab+REST APIs, https://github.com/pooyanjamshidi/autoscaling-bigdata

LD [S10] **TL4CO**, *A Machine Learning tool for finding the optimum configuration of Big Data systems by transferring the learning from other system versions in DevOps context*, Matlab+Java, https://github.com/dice-project/DICE-Configuration-TL4CO

★LD [S9] **BO4CO**, *A Machine Learning tool for finding the optimum configuration of Big Data systems*, Matlab+Python, https://github.com/dice-project/DICE-Configuration-BO4CO

LD [S8] **ElasticBench**, *A cloud application framework to plug-in auto-scaling logic and experimentally evaluate controllers in a feedback control loop on platform as a service environment on Microsoft Azure*, .NET, https://github.com/pooyanjamshidi/ElasticBench

CC [S7] **spark-suite**, *A suite for automated configuration testing, automated topology deployment and a benchmarking tool for Apache Spark*, Java, https://github.com/pooyanjamshidi/spark-suite

CC [S6] **OSTIA**, *A parser to elicit and represent Storm topologies by reverse engineering Storm-based programs*, Ruby, https://github.com/maelstromdat/OSTIA

LD [S5] **pong-engine**, *An engine that runs pong games on Matlab and paddles are controlled by reinforcement learner agents. I implemented this piece of software for a reinforcement learning course*, Matlab, https://github.com/pooyanjamshidi/pong-engine

CC [S4] **MDLoad**, *MDload is a model-driven workload generation tool that automatically generates requests to a web application by simulating a set of users*, Java+Matlab, https://github.com/imperial-modaclouds?query=modaclouds-mdload

LD [S3] **Fuzzy-Q-Learning**, *An implementation of Fuzzy Q-Learning for making cloud auto-scaling more intelligent through online policy learning*, Matlab, https://github.com/pooyanjamshidi/Fuzzy-Q-Learning

LD [S2] **RobusT2Scale**, *A cloud auto-scaler based on fuzzy reasoning*, Matlab, https://github.com/pooyanjamshidi/RobusT2Scale

LD [S1] **ASIM**, *A program that automatically identifies services out of business processes*, Java, https://github.com/pooyanjamshidi/ASIM

Data

I release the data that I collect for my research to the public community for replication.

[D3] **ATHENA**, *The dataset (30GB) associated with ATHENA—a framework based on Diverse Weak Defenses for building Adversarial Defense.*, https://zenodo.org/record/4141383

[D2] **ASE 2017**, *Transfer Learning for Performance Modeling of Configurable Systems: An Exploratory Analysis*, **Subject systems**: SaC, SQLite, SPEAR, X264, https://github.com/pooyanjamshidi/ase17

[D1] **MASCOTS 2016**, *An Uncertainty-Aware Approach to Optimal Configuration of Stream Processing Systems*, **Subject systems**: Apache Storm, Apache Spark, Apache Hadoop, Apache Cassandra, https://zenodo.org/record/56238

Services

Mentoship

URM Mentor — **Mentor for underrepresented minority**, *I serve as a mentor for underrepresented minority (URM) students in an R25 NIH/NIAID funded program*, Big Data Health Science Emerging Scholar (e-Scholar) Program for Infectious Diseases, R25AI172761-01; MPI: Xiaoming Li & Jiajia Zhang, total award: $1.565m, 06/23-05/28.
https://reporter.nih.gov/search/RwlKQSggdUOsj4SbB9iCiA/project-details/10554786

Top Scholar Mentor — **Mentor for USC's Top Scholars**, *I have been actively serving as a mentor for USC's Top Scholars:*

9/2021- **Chance Storey**

9/2019- **Hadley Blalock**

Co-Founder

JSys — **Co-founded JSys (Journal of Systems Research)**, *a new diamond open-access journal with Vijay Chidambaram, Neeraja Yadwadkar, Ivo Jimenez, and Romain Jacob*, https://www.jsys.org/

Gamecock Robotics — **Mentoring the Gamecock Robotics team**, *a student-led team for a competitive robotics league called Vex Robotics*, http://gamecockrobotics.github.io/

SPEC RG — **Electecd as the vice chair of SPEC RG Predictive Data Analytics**, *https://research.spec.org/working-groups/rg-predictive-data-analytics/*

Editorial

JSys — **Co-Editor at JSys (Journal of Systems Research)**, *Two Areas: Computer Architecture (with Devashree Tripathy at IIT Bhubaneswar) and Configuration Management for Systems (with Tianyin Xu at UIUC)*, https://www.jsys.org/board

IEEE Software — **IEEE Software Special Issue on Microservices, Guest editor**, *Co-edited with James Lewis (ThoughtWorks), Stefan Tilkov (innoQ), Claus Pahl, and Nabor Mendonça, This special issue attracted 26 submissions, a record number in IEEE Software*, https://www.computer.org/software-magazine/2017/02/10/microservices-call-for-papers/

Program Co-Chair

SEAMS 2023 — The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Program Co-Chair with Valerie Issarny (update: Program Co-Chair with Radu Calinescu and Raffaela Mirandola).

Area Chair

AutoML-Conf 2023 — 2nd International Conference on Automated Machine Learning.

AutoML-Conf 2022 — 1st International Conference on Automated Machine Learning.

Co-organizer

SEAMS 2023 — The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, Program Co-Chair.

MLArchSys at ISCA 2023 — ML for Computer Architecture and Systems, Co-Organizer and PC.

SEAMS 2022	The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, social media chair.
ENSEMBLE 2019	2nd International Workshop on Ensemble-based Software Engineering for Modern Computing Platforms (co-located with ESEC/FSE 2019), co-chair.
SEAMS 2017	The International Symposium on Software Engineering for Adaptive and Self-Managing Systems, publicity and proceedings chair (Co-organized with David Garlan, Bashar Nuseibeh, Javier Camára, and Nicolás D'Ippolito).
CloudWays 2017	International Workshop on Cloud Adoption and Migration, Workshop co-chair (Co-organized with Claus Pahl, and Nabor Mendonça).
Dagstuhl 2016	Software Performance Engineering in the DevOps World, Co-organized the seminar with Andre van Hoorn, Philipp Leitner, and Ingo Weber.
CloudWays 2016	International Workshop on Cloud Adoption and Migration, Workshop co-chair.
CloudWays 2015	International Workshop on Cloud Adoption and Migration, Workshop co-chair.

Program Committees (Conferences)

ICSE 2023	International Conference on Software Engineering
ICPE 2023	International Conference on Performance Engineering
ASE 2022	International Conference on Automated Software Engineering - Artifacts Track
SEAMS 2022	Software Engineering for Adaptive and Self-Managing Systems
ICPE 2022	International Conference on Performance Engineering
ICSE/NIER 2022	International Conference on Software Engineering
ASE 2021	International Conference on Automated Software Engineering - Artifacts Track
AISTATS 2021	International Conference on Artificial Intelligence and Statistics
SEAMS 2021	16th Software Engineering for Adaptive and Self-Managing Systems
ICSA 2021	International Conference on Software Architecture
ICPE 2021	International Conference on Performance Engineering
XP 2020	International Conference on Agile Software Development
ICSE 2020	International Conference on Software Engineering
SEAMS 2020	15th Software Engineering for Adaptive and Self-Managing Systems
ICPE 2020	International Conference on Performance Engineering
VaMoS 2020	International Conference on Variability Modeling of Software-Intensive Systems
ECSA 2019	13th European Conference on Software Architecture
SATURN 2019	SEI Architecture User Network (SATURN) Conference
ICSA 2019	International Conference on Software Architecture (Tool Track)
Microservices 2019	International Conference on Microservices
SEAMS 2019	14th Software Engineering for Adaptive and Self-Managing Systems
ICSOC 2018	16th International Conference on Service-Oriented Computing
ECSA 2018	12th European Conference on Software Architecture
ICDCS 2018	38th International Conference on Distributed Computing Systems
SEAMS 2018	13th Software Engineering for Adaptive and Self-Managing Systems
ECSA 2017	11th European Conference on Software Architecture
SEAMS 2017	12th Software Engineering for Adaptive and Self-Managing Systems
EUSPN 2017	8th Emerging Ubiquitous Systems and Pervasive Networks

SIGMOD 2016	ACM SIGMOD 2016 Reproducibility
SEAMS 2016	11th Software Engineering for Adaptive and Self-Managing Systems
EUSPN 2016	7th Emerging Ubiquitous Systems and Pervasive Networks
ICSOFT 2016	13th International Conference on Software Technologies
ICSOFT 2015	12th International Conference on Software Technologies

Program Committees (Workshops)

WAM at ESEC/FSE'23	1st International Workshop on Application Modernization
ASSYST at ISCA'23	Architecture and System Support for Transformer Models
AMP 2021	International Workshop on Agility with Microservices Programming
AKSAS 2018	International Workshop on Architectural Knowledge for Self-Adaptive Systems
MLMH 2018	KDD Workshop on Machine Learning for Medicine and Healthcare
AMS 2018	International Workshop on Architecting with MicroServices
SQUADE 2018	International Workshop on Software Qualities and their Dependencies
LTB 2018	Load Testing and Benchmarking of Software Systems
ASBDA 2017	International Workshop on Autonomic Systems for Big Data Analytics
AMS 2017	International Workshop on Architecting with MicroServices
QUDOS 2017	International Workshop on Quality-Aware DevOps
LTB 2017	Load Testing and Benchmarking of Software Systems
QUORS 2017	International COMPSAC Workshop on Quality Oriented Reuse of Software
LTB 2016	Load Testing and Benchmarking of Software Systems
QUORS 2016	International COMPSAC Workshop on Quality Oriented Reuse of Software

Journal Reviews

Brackets indicate the number of papers I reviewed (excluding revisions).

TSE (15)	IEEE Transactions on Software Engineering
TOSEM (15)	ACM Transactions on Software Engineering and Methodology
TAAS (10)	ACM Transactions on Autonomous and Adaptive Systems
Software (7)	IEEE Software
ROBOT (3)	Elsevier Robotics and Autonomous Systems
TSC (3)	IEEE Transactions on Service Computing
TCC (3)	IEEE Transactions on Cloud Computing
SPI (3)	Wiley Software Process: Improvement and Practice
IST (3)	Elsevier Information and Software Technology
Computing (3)	Springer Computing
JPDC (3)	Journal of Parallel and Distributed Computing
SoSyM (2)	Springer Software & Systems Modeling
Oxf Comp Jrnl (2)	Oxford Academic Computer Journal
JSEP (2)	Wiley Journal of Software: Evolution and Process
SPE (2)	Wiley Software: Practice and Experience
Cloud (2)	IEEE Cloud Computing
IET Software (2)	IET Software
PLOS ONE (2)	PLOS ONE
ToMPECS (1)	ACM Trans. on Modeling and Performance Evaluation of Computing Systems
CSUR (1)	ACM Computing Surveys
Micro (1)	IEEE Micro
ASC (1)	Elsevier Applied Soft Computing
Computer (1)	IEEE Computer
Computing (1)	IEEE Internet Computing
JNCA (1)	Elsevier Journal of Network and Computer Applications
TNSM (1)	IEEE Transactions on Network and Service Management
JCST (1)	Springer Journal of Computer Science and Technology
JCC (1)	Springer Journal of Cloud Computing
SOCA (1)	Springer Service Oriented Computing and Applications
JSA (1)	Elsevier Journal of Systems Architecture
JBCS (1)	Springer Journal of the Brazilian Computer Society
JSS (1)	Elsevier Journal of Systems and Software
JWE (1)	Journal of Web Engineering
IBM (1)	IBM Journal of Research and Development
Computers (1)	MDPI Computers
Entropy (1)	MDPI Entropy
Supercomp. (1)	Springer Journal of Supercomputing
EMSE (1)	Springer Empirical Software Engineering
CACM (1)	Communications of the ACM

AIJ (1)	Elsevier Artificial Intelligence Journal
Neurocomp (1)	Elsevier Neurocomputing
Access (1)	IEEE Access
Neurocomp (1)	Frontiers in Epidemiology
TECS (1)	ACM Transactions on Embedded Computing Systems

Grant proposal review

2023	Swiss National Science Foundation (SNSF)
2022	German Research Foundation, Deutsche Forschungsgemeinschaft (DFG)
2021	National Science Foundation (NSF) panel (CISE/CNS)
2021	National Science Foundation (NSF) panel (CISE/SHF)
2020	National Science Foundation (NSF) panel (CISE/IIS)
2019	Dutch Research Council (NWO)
2018	Canadian Science Fund (FRQnet)
2018	Austrian Science Fund (FWF)
2016	Dutch Technology Foundation (STW)

Other reviews

2016	Elsevier book proposal review

Industry Services

SPEC	I have been actively collaborating with the DevOps RG group for developing a DevOps framework by consolidating tools to better integrate performance monitoring and architectural refactoring.
Intel and Microsoft	During my postdoctoral research in IC4 (Irish cloud center with 40+ industry members), I was actively collaborating with Giovani Estrada and Chris Woods from Intel and Niall Moran from Microsoft for developing auto-scaling mechanisms for OpenStack and Azure platforms, see [**C15, C14, C18**].
IPMA Project Management Excellence	My responsibility was to *assess* the quality of the projects submitted for the "National Project Management Excellence Award" and to judge their excellence by exploiting the IPMA Project Excellence Model. The assessment process included individual assessments, consensus meetings, site visits and report writing.
Apache	I am active in the Apache Storm community contributing to the auto-scaling feature of the framework: https://issues.apache.org/jira/browse/STORM-594
Imperial Consultants	I was a research consultant on Big Data and Machine Learning in Imperial Consultants, a self-funding, wholly owned company of Imperial College London.

Outreach Activities

Outreach	High school student mentor for CREST Awards 2015, Project title: "Can a program beat the Turing test?", Queens Park School, UK.
Outreach	Young Scientists Exhibition 2013, Lero stands, Dublin, Ireland, 2013.

Membership in Technical Communities

2019–	AAAI
2015–	SPEC RG DevOps Performance Group
2011–	IEEE, ACM, ACM SIGSOFT

Immigration Status

US Permanent Resident (Green Card holder since April 2020).

References

Christian Kästner
Postdoc Advisor
Assistant Professor, Carnegie Mellon University, USA.
+1 412-268-5254 kaestner@cs.cmu.edu
https://www.cs.cmu.edu/~ckaestne/

David Garlan
Mentor
Professor, Carnegie Mellon University, PA, USA.
+1 412-268-5056 garlan@cs.cmu.edu
https://www.cs.cmu.edu/~garlan/

Claus Pahl
Ph.D. Advisor
Associate Professor, Free University of Bozen-Bolzano, Italy.
+39 0471-016-177 Claus.Pahl@unibz.it
https://www.inf.unibz.it/~cpahl/

Ladan Tahvildari
Mentor
Professor, University of Waterloo, Canada.
+1 519-888-4567 ladan.tahvildari@uwaterloo.ca
https://uwaterloo.ca/electrical-computer-engineering/profile/ltahvild

Isil Dillig
Mentor
Associate Professor, UT Austin, USA.
+1 (512) 471-9794 isil@cs.utexas.edu
https://www.cs.utexas.edu/~isil/

Jason O'Kane
Mentor
Professor, Texas A&M, Texas, USA.
+39 0471-016-177 Jokane@cse.sc.edu
https://www.cse.sc.edu/~jokane/

Homay Valafar
Mentor
Professor, University of South Carolina, SC, USA.
+39 0471-016-177 homayoun@cec.sc.edu
https://ifestos.cse.sc.edu/